COURSE REPORT

DEVELOPMENT OF THE PYTHON BASED SURVEY UNIT CONVERSION AND MEASUREMENT CALCULATOR USING PYCHARM

1. Introduction

Surveying involves measuring distances, areas and angles in order to determine the positions, sizes and shapes of land features and properties. Measurements collected in the field or used during mapping may be expressed in different units. A surveyor may therefore need to convert metres into kilometres, centimetres, millimetres, feet or yards, change areas between hectares and square metres, or convert angles between degrees and radians.

Carrying out these conversions and basic calculations manually can take time, especially when several measurements have to be processed. A computer-based calculator can make the work easier by accepting values from the user, applying the appropriate mathematical formula and displaying the result. This project developed a basic Python-based survey calculator using the PyCharm integrated development environment (IDE).

2. Description of the Problem

The main problem addressed by the project is the time and possibility of arithmetic errors involved in manually converting surveying measurements and performing simple survey calculations. Survey students and practitioners may work with measurements expressed in different units. If a value is converted using the wrong factor, or if a calculation is performed incorrectly, the resulting distance or area may be inaccurate.

Manual calculations can also become repetitive when many measurements are involved. This may slow down work and make it harder to check results consistently. The project therefore responds to the need for a simple digital tool that brings common unit conversions and selected measurement calculations together in one program.

3. Objectives of the Project

The overall objective was to develop a Python program that converts common surveying measurements and performs simple calculations accurately.

- To create a calculator that converts length measurements, including metres, kilometres, centimetres, millimetres, feet and yards.
- To provide area conversions between hectares, square metres and acres.
- To convert angular measurements from degrees to radians and from radians to degrees.
- To calculate the horizontal distance between two survey points using their coordinates.
- To calculate the area of a rectangular plot from its length and width.
- To test the calculator using sample surveying values and compare the results with the expected mathematical answers.

4. Methodology

The project followed a practical, step-by-step development approach. The requirements were identified first, after which the development environment was prepared, the Python file was created, the program logic was written, and sample calculations were used to test the calculator.

4.1 Software and Development Tools

- Python 3: the programming language used to write the calculator and perform calculations.
- PyCharm: the integrated development environment used to create, edit, save and run the Python program.
- Python math module: used for mathematical operations such as square roots and conversion between degrees and radians.

4.2 Project Setup and Implementation

1. PyCharm was opened and a new project named SurveyCalculator was created.
2. A valid Python interpreter and a new virtual environment were selected during project setup.
3. A Python source file named survey_calculator.py was created inside the project.
4. The program logic was entered in the Python file and saved using the Save All command or Ctrl + S.
5. The file was run from PyCharm using the Run command or the green Run button. The program's menu and results were intended to appear in the Run window.

4.3 Main Operations and Calculations

The calculator was designed around a menu that allows a user to select the required operation, enter a measurement and view the calculated result. The main operations and mathematical approaches are summarised below.

Operation	**Calculation / conversion approach**
Length conversion	Uses conversion factors, for example metres to kilometres = metres ÷ 1,000; metres to centimetres = metres × 100.
Area conversion	Hectares to square metres = hectares × 10,000; hectares to acres = hectares × 2.4710538147.
Angle conversion	Uses the Python math functions radians() and degrees().
Distance between points	Uses the 2D coordinate distance formula: $D = \sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]}$.
Rectangular plot area	Area = length × width.

4.4 Input, Processing and Output

The user selects an operation from the menu and enters the required numerical values. The input() function receives the entries, while float() converts numerical entries into decimal values that can be used in calculations. Conditional statements such as if, elif and else select the correct operation. The result is then displayed using print(). Reusable functions defined with def organise the different calculator operations. The guide also identifies while True as a way of keeping the menu active until the user chooses Exit, and try and except as a way of handling invalid numerical input.

5. Results and Discussion

The calculator was intended to be checked using sample surveying examples. The supplied project guide gives four test cases and their expected mathematical results. These results are presented below. They are the expected results stated in the guide; no separate run-output screenshots or independent test log were included in the supplied material.

Test	Input values	Calculation	Expected result
1. Length conversion	500 metres	500 ÷ 1,000	0.5000 kilometres
2. Area conversion	2 hectares	2 × 10,000	20,000.0000 square metres
3. Distance between survey points	A (100, 200); B (400, 600)	$\sqrt{[(400-100)^2 + (600-200)^2]}$	500.000 metres
4. Rectangular plot area	Length = 100 m; width = 50 m	100 × 50	5,000.000 m² (0.5000 ha)

5.1 Interpretation of the Results

The first test shows that 500 metres is equal to 0.5 kilometres. This confirms the relationship that one kilometre contains 1,000 metres. The second test shows that 2 hectares are equal to 20,000 square metres because one hectare contains 10,000 square metres.

For the coordinate-distance test, the difference between the eastings is 300 metres and the difference between the northings is 400 metres. Applying the distance formula gives $\sqrt{(300^2 + 400^2)} = \sqrt{250,000} = 500$ metres. This is the straight-line horizontal distance between the two points in the stated coordinate system.

For the rectangular plot, multiplying 100 metres by 50 metres gives 5,000 square metres. Since one hectare is 10,000 square metres, the same area is 0.5 hectares. The results demonstrate how the calculator can support basic surveying conversions and calculations when the correct inputs and units are supplied.

6. Challenges Encountered

The project guide identifies several common problems that may be encountered while developing and running the calculator. The challenges below are based on those troubleshooting points; they should be understood as possible implementation challenges rather than a claim that every error occurred during the project.

- Indentation errors: Python uses indentation to show which statements belong inside functions and conditional blocks. Inconsistent spacing can prevent the program from running.
- Syntax errors: Missing colons, brackets or quotation marks, or typing mistakes in statements, can cause the interpreter to stop the program.
- Python interpreter configuration: PyCharm may fail to run the program if a valid Python interpreter has not been installed or selected in the project settings.
- Running the wrong file: If a different file is selected when Run is clicked, the expected calculator menu may not appear.
- Invalid numerical input: Entering words instead of numbers can cause a ValueError when the program attempts to convert the input to a float.
- Incorrect conversion selection or units: A user may obtain an unexpected answer after choosing the wrong menu option or entering a value in a unit different from the one requested.

7. Limitations and Precautions

The distance calculation has an important limitation. It calculates a two-dimensional horizontal distance using coordinate values. The two points must be in the same suitable projected coordinate system, and both coordinate axes must be expressed in metres. The function does not calculate slope distance, elevation difference or geodesic distance over the Earth's surface.

For this reason, survey users should check the units and coordinate reference system before entering coordinates. They should also confirm the accuracy and suitability of the measurements. The calculator is intended for basic conversions and simple computations; it does not replace professional surveying procedures, field checks or specialised surveying software.

8. Conclusion

The project focused on developing a Python-based survey unit conversion and measurement calculator using PyCharm. Its purpose was to reduce the time spent on repetitive manual conversions and basic calculations, while helping to minimise arithmetic mistakes. The planned calculator brings together length, area and angle conversions, coordinate-based horizontal distance calculation and rectangular plot-area calculation.

The sample test cases provided in the project guide produced clear expected results: 500 metres equals 0.5000 kilometres, 2 hectares equal 20,000 square metres, the distance between the sample coordinate points is 500 metres, and a rectangular plot measuring 100 metres by 50 metres has an area of 5,000 square metres or 0.5 hectares. These examples show the usefulness of a simple programming tool in supporting surveying computations.

In conclusion, the project demonstrates how programming can be applied in surveying to make routine measurement work more organised and efficient. However, correct results depend on selecting the right operation, entering valid values and using suitable units and coordinate systems.

9. Recommendations

- Test every menu option with several values, including decimal values, before relying on the calculator.
- Add clear input validation and error messages so that incorrect entries can be corrected without terminating the program.
- Display the source unit and destination unit clearly for every conversion.
- Keep a record of test inputs and actual program outputs, and include screenshots from PyCharm in the final submission.
- For coordinate-distance work, confirm the coordinate reference system and units before using the calculated distance.

Source Used

Pycharm-exported.pdf, “Development of a Python-Based Survey Unit Conversion and Measurement Calculator Using PyCharm,” CS 110 – How to Program coursework guide, six pages. The report is based on the project description, objectives, implementation guidance, test examples and troubleshooting information in the supplied document.